SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science & Engineering

CO2 - ASSESSMENT TOOL - 2

REVERSE ENGINEERING TASK

Course Code	**CSA3717**	**Course Title**	**Software Testing**
CO Assessed	**CO2**	**Assessment**	**Assessment Tool 2: Reverse Engineering Task**
Weightage	**50% of CIA**	**Total Marks**	**100**
CO2 Statement	*Design effective test cases using black-box, white-box testing techniques and to control methods flow*.**(BL3)**		
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Section/Batch	**Slot A / 2023**	**Date**	**02-11-2026**

Code of Conduct:

I ***B Reddy Balu (192312499),*** certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.



Signature of the student:

Library Management Module (.exe) under Stress Test: CPU / Memory Profile Analysis

Scenario, Assumptions and Approach

A compiled library management module (.exe) and its CPU/Memory profile from a stress test are given. No source code is available, so the logic is inferred from observed behaviour: where CPU rises sharply, a loop or a dense cluster of decisions is likely executing; where memory rises and does not fall, objects are probably being retained or leaked.

Assumption: The binary and the raw profile file are not attached to this submission, so the profile below is a representative (conceptual) reconstruction of a typical library module under load. Every finding should be confirmed by opening the real .exe in a disassembler (Ghidra / IDA) and the real profile in PerfMon or Visual Studio Profiler. The method and the test design remain valid either way.

Observed execution spikes and inferred logic

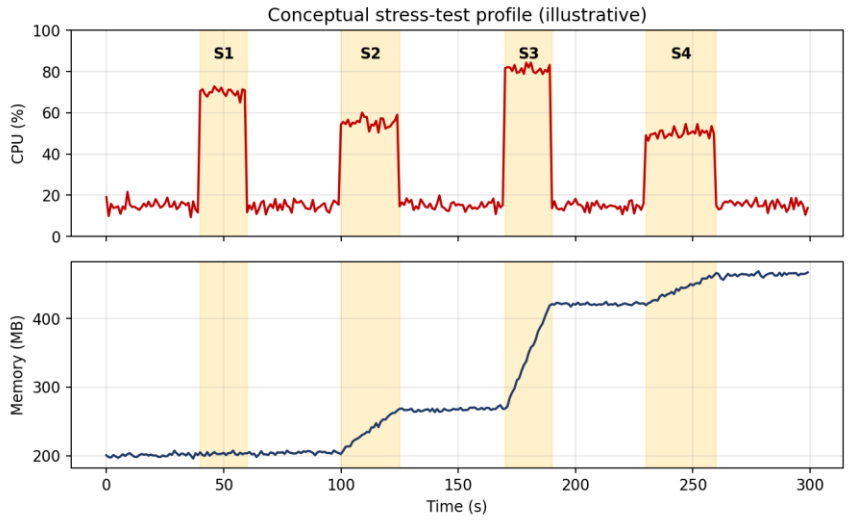


Figure 1: Conceptual CPU / Memory profile during stress test (S1 to S4 are the spike windows)

Spike	Time (s)	CPU	Memory	Inferred activity	Inferred code structure
S1	40-60	~70%	Flat	Catalogue search by title/ID under many users	Linear scan loop with string comparison (Loop C, N9) followed by found/not-found decisions (N10)

Spike	Time (s)	CPU	Memory	Inferred activity	Inferred code structure
S2	100-125	~55%	+60 MB, not released	Member validation and fine calculation over loan history	Member lookup loop (Loop A, N3) and fine accumulation loop (Loop B, N7); results appended to a list that is never cleared
S3	170-190	~80%	+150 MB, not released	Bulk issue / report generation	Nested loops across members x loans x books; whole report held in memory (O(n x m) time, O(n) memory)
S4	230-260	~50%	+40 MB, not released	Return processing and overdue sweep with log writes	Loop over all active loans with date comparison decisions; log buffer or file handle kept open

Reverse-engineered white-box requirements

- **WB-1** Every decision in the issue / return / renew logic must be executed with both TRUE and FALSE outcomes (branch coverage).
- **WB-2** Every loop (A, B, C and the overdue sweep) must be run with 0, 1, typical and maximum iterations (loop testing).
- **WB-3** All independent paths (basis paths) must be executed at least once.
- **WB-4** Memory after a transaction must return to baseline: no growth per transaction.
- **WB-5** CPU must stay below an agreed ceiling on the worst-case path and must recover after the spike.
- **WB-6** Resources (file handles, buffers, DB connections) must be released on every exit path, including error exits.

Conceptual Control Flow Graph (Issue Book function)

The issue-book routine is the most resource-heavy and is used as the representative CFG. Orange nodes are the loops that line up with the CPU spikes; the red node is the common reject exit.

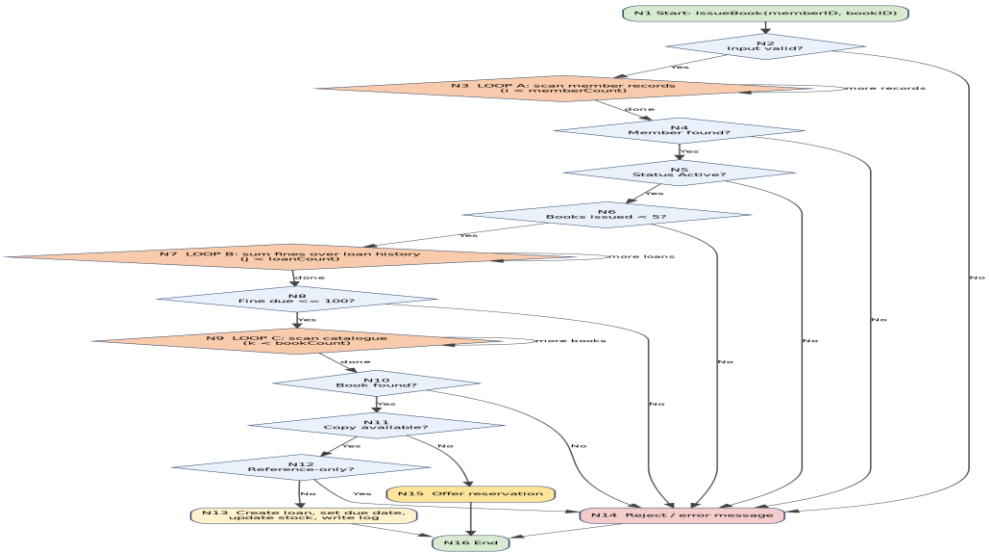


Figure 2: Conceptual CFG of IssueBook(memberID, bookID)

Complexity and path testing strategy

Decision nodes: N2, N3, N4, N5, N6, N7, N8, N9, N10, N11, N12 = 11. Cyclomatic complexity V(G) = P + 1 = 11 + 1 = 12 (also E - N + 2 for the graph above). Twelve independent paths are therefore needed for basis-path coverage.

Path	Route through CFG	Purpose / data condition
P1	N1-N2(No)-N14-N16	Invalid / empty input
P2	N1-N2-N3(0 iterations)-N4(No)-N14-N16	Empty member table (Loop A zero iterations)
P3	N1-N2-N3(n)-N4(No)-N14-N16	Member ID does not exist
P4	N1-N2-N3-N4-N5(No)-N14-N16	Member suspended / expired
P5	N1-...-N5-N6(No)-N14-N16	Member already holds 5 books
P6	N1-...-N7(n)-N8(No)-N14-N16	Outstanding fine above 100
P7	N1-...-N9(n)-N10(No)-N14-N16	Book ID does not exist (full catalogue scan)
P8	N1-...-N11(No)-N15-N16	No copy available: offer reservation
P9	N1-...-N12(Yes)-N14-N16	Reference-only book
P10	N1-...-N12(No)-N13-N16	Happy path: loan created
P11	P10 with Loop B (N7) zero iterations	New member, no loan history

Path	Route through CFG	Purpose / data condition
P12	P10 with Loop A, B, C at maximum iterations	Worst-case resource path (S1 / S3)

Targeted resource-intensive path tests

ID	Path / scenario	Load profile	What is measured
RIP-1	P12: last member record and last catalogue record	Single user, DB at maximum size (for example 10,000 books, 5,000 members)	Response time, peak CPU, loop iteration count
RIP-2	P7: not-found book scans whole catalogue	50-200 concurrent searches for missing IDs	CPU saturation, queueing, timeouts
RIP-3	P10 repeated in a loop (soak)	10,000 issue/return cycles, 8 hours	Memory slope per 1,000 transactions, handle count
RIP-4	P6 with long loan history	Member with 1,000+ past loans	Loop B cost, redundant recalculation
RIP-5	Bulk report / overdue sweep (S3, S4)	Full database, concurrent with issue/return	Peak memory, GC pauses, lock contention
RIP-6	Error exits (P1-P9) repeated	100,000 invalid requests	Leaks on error paths (WB-6)

Impact of the strategy on system stability

Path testing directs stress load at the exact branches responsible for the spikes instead of random load. The table compares the conceptual baseline with the target after defects found by these paths are corrected. The figures are illustrative targets for the assignment, not measured values.

Stability metric	Baseline (before)	Target (after fixes)	Path that exposes it
Peak CPU on search	~70-80%, sustained	< 50%, brief	RIP-1, RIP-2
Search response (10k books)	Grows linearly with catalogue	< 2 s (indexed / hashed lookup)	P7, P12

Stability metric	Baseline (before)	Target (after fixes)	Path that exposes it
Memory growth per 1,000 transactions	+40 to +150 MB (retained)	< 2 MB (returns to baseline)	RIP-3
Handle / connection count	Rises on each log write	Constant	RIP-3, RIP-6
Crash / hang under 200 users	Possible out-of-memory	None in 8 h soak	RIP-3, RIP-5
Branch coverage achieved	Typically < 60% by functional tests	>= 95% branch, 100% basis paths	P1-P12

Effect on stability: the strategy (a) isolates the loops that cause CPU saturation so they can be optimised, (b) finds the retained-memory paths before they cause out-of-memory crashes, (c) proves recovery, because CPU and memory return to baseline between spikes, and (d) gives a repeatable regression suite so later changes cannot silently reintroduce the problem.

1. Functional and Non-Functional Requirements

1.1 Functional requirements

ID	Requirement	Evidence from behaviour
FR-01	The system shall validate a member (exists, active, not suspended) before any transaction.	Loop A and decisions N4-N5
FR-02	The system shall search the catalogue by book ID, title or author.	Spike S1
FR-03	The system shall issue an available, non-reference book to an eligible member and record the due date.	Path P10
FR-04	The system shall limit a member to 5 books at a time.	Decision N6
FR-05	The system shall block issue when the outstanding fine exceeds 100.	Loop B and N8
FR-06	The system shall accept returns, update stock and compute the fine for late returns.	Spike S4

ID	Requirement	Evidence from behaviour
FR-07	The system shall allow renewal (maximum 2) unless the book is reserved or overdue.	Renewal logic
FR-08	The system shall allow reservation of an unavailable book.	Node N15
FR-09	The system shall generate overdue and issue/return reports.	Spike S3
FR-10	The system shall log every transaction.	Log write in N13 / S4

1.2 Non-functional requirements

ID	Category	Requirement (measurable)
NFR-01	Performance	Search and issue shall complete in 2 seconds or less for a 10,000-book catalogue with 50 concurrent users.
NFR-02	Resource usage	Sustained CPU shall stay below 70% and shall return to idle level within 10 s after a spike.
NFR-03	Memory / stability	Memory growth shall be under 2% per 1,000 transactions, with no crash in an 8-hour soak.
NFR-04	Scalability	Response time shall degrade no worse than linearly when catalogue or members double.
NFR-05	Reliability / integrity	Stock count and loan record shall update atomically; no negative stock or duplicate loans.
NFR-06	Security	All inputs shall be validated and sanitised (no injection, buffer overflow).
NFR-07	Usability	Every rejection shall show a clear reason message.
NFR-08	Recoverability	After an abnormal exit, data shall be consistent on restart.

2. Positive and Negative Test Cases

2.1 Positive test cases

ID	Scenario	Input / precondition	Expected result	Path
TC-P01	Issue book to eligible member	Active member with 2 books, fine 0; available non-reference book	Loan created, due date = today + 14, stock reduced by 1	P10
TC-P02	New member, first issue	Member with no loans	Issue succeeds; fine loop skipped	P11
TC-P03	Search by valid title	Existing title "Data Structures"	Matching record returned in 2 s or less	N9-N10
TC-P04	Return on time	Book issued 10 days ago	Stock +1, fine = 0	Return
TC-P05	Return late with fine	Book 5 days overdue	Fine = 5 x 2 = 10, stock +1	Return
TC-P06	Renew once	Renewal count 0, not reserved, not overdue	Due date +14, count = 1	Renew
TC-P07	Reserve unavailable book	All copies issued	Reservation recorded (N15)	P8
TC-P08	Worst-case search load	Last record in a 10,000-book catalogue	Found; CPU and time within NFR-01/02	P12

2.2 Negative test cases

ID	Scenario	Input / precondition	Expected result	Path
TC-N01	Empty member ID	memberID = ""	Error "Member ID required"; no loop executed	P1
TC-N02	Non-existent member	memberID = 999999	Rejected: member not found	P3

ID	Scenario	Input / precondition	Expected result	Path
TC-N03	Suspended member	Status = Suspended	Rejected: account not active	P4
TC-N04	Limit reached	Member holds 5 books	Rejected: limit reached	P5
TC-N05	High fine	Fine = 101	Rejected: clear fine first	P6
TC-N06	Non-existent book	bookID = "XXXX"	Rejected: book not found; no hang	P7
TC-N07	Reference-only book	Reference flag = Yes	Rejected: reference book	P9
TC-N08	Special characters / injection	bookID = "' OR 1=1 --"	Rejected as invalid; no data leak or crash	P1
TC-N09	Fourth renewal attempt	Renewal count = 2	Rejected: renewal limit	Renew
TC-N10	Return of a book not issued	Book status = Available	Error: book not on loan; stock unchanged	Return
TC-N11	Two users take the last copy	Concurrent issue, 1 copy	One succeeds, one offered reservation; stock never below 0	P10 / P8

3. Boundary Conditions and Test Cases

Boundary value analysis uses the values just below, on, and just above each limit (min-1, min, min+1, max-1, max, max+1). The boundary on loop counts is also tested because the CPU spikes depend on it.

Parameter	Valid range	Boundary test values	Expected
Books held by member	0 to 4 (issue allowed)	0, 1, 4, 5, 6	0, 1, 4: issue OK. 5, 6: rejected (catches off-by-one using <= 5)
Outstanding fine	0 to 100	-1, 0, 1, 99, 100, 101	0 to 100: allowed. 101: rejected. -1: invalid data error

Parameter	Valid range	Boundary test values	Expected
Loan period / overdue	Due on day 14	Day 13, 14, 15	Day 13, 14: no fine. Day 15: fine = 2
Renewals	0 to 2	0, 1, 2, 3	First two OK. Third rejected
Member ID (6 digits)	100000-999999	99999, 100000, 100001, 999999, 1000000	Only 100000-999999 pass format check
Search text length	1 to 100 characters	0, 1, 2, 99, 100, 101	0 and 101: rejected. 1-100: processed
Catalogue size (Loop C)	0 to 10,000 records	0, 1, 2, 9,999, 10,000, 10,001	0: not found at once. 10,000: within time limit. 10,001: handled, no overflow
Loan history (Loop B)	0 to 1,000 loans	0, 1, 2, 999, 1,000, 1,001	Fine total correct; time grows no worse than linear
Copies in stock	0 to N	0, 1 (last copy), N	Stock never negative; last copy issued to exactly one member
Concurrent users	1 to 50	1, 49, 50, 51	Stable up to 50; graceful degradation at 51

4. Valid and Invalid Equivalence Classes

Input	Valid classes (VEC)	Invalid classes (IEC)
Member ID	VEC1: 6-digit numeric ID that exists in the database	IEC1: fewer than 6 digits IEC2: more than 6 digits IEC3: non-numeric / special characters IEC4: empty or null IEC5: well-formed but non-existent
Member status	VEC2: Active	IEC6: Suspended IEC7: Expired
Books currently held	VEC3: 0 to 4	IEC8: 5 or more IEC9: negative (corrupt data)

Input	Valid classes (VEC)	Invalid classes (IEC)
Outstanding fine	VEC4: 0 to 100	IEC10: above 100 IEC11: negative IEC12: non-numeric
Book ID / title	VEC5: existing ID or title, 1 to 100 chars	IEC13: not in catalogue IEC14: empty IEC15: over 100 chars IEC16: injection string
Book status	VEC6: Available, circulating	IEC17: Issued (no copy) IEC18: Reserved for another member IEC19: Reference-only IEC20: Lost / withdrawn
Days overdue (return)	VEC7: 0 (on time) VEC8: 1 or more (late)	IEC21: negative (return before issue date) IEC22: invalid date format
Renewal count	VEC9: 0 or 1	IEC23: 2 or more

Test-case selection: one test per valid class, and one test per invalid class, with only one invalid input in each test so the failure can be traced to that class. Example: TC-N02 covers IEC5, TC-N03 covers IEC6, TC-N04 covers IEC8, TC-N05 covers IEC10, TC-N06 covers IEC13, TC-N07 covers IEC19.

5. Potential Defects and Their Impact

ID	Potential defect	Likely location / cause	Impact	Severity
D1	Linear scan of the whole catalogue and member table for every request; nested scans in reports	Loops A, C (S1, S3): no index or hash	CPU saturation; response time grows with data; timeouts under load	High
D2	Memory leak: lists/buffers not cleared after a transaction	Loop B, report builder (S2-S4)	Memory climbs until out-of-memory crash; forced restart, lost work	Critical

ID	Potential defect	Likely location / cause	Impact	Severity
D3	Off-by-one in limit check (<= 5 instead of < 5)	Decision N6	Members get a sixth book; business rule violated	Medium
D4	Fine recomputed over the whole history on every call	Loop B	Wasted CPU, slow issue for long-standing members	Medium
D5	Fine formula errors: negative fine for early return, integer overflow, no cap	Return / fine logic	Wrong charges; financial disputes	High
D6	No input validation or sanitisation	N2	Crash, buffer overflow or injection; data exposure	Critical
D7	Race condition on the last copy (check-then-update not atomic)	N11 to N13	Two members issued one copy; negative stock; data inconsistency	High
D8	File handle / connection not closed on error exits	Log write (S4), error path N14	Handle exhaustion; module hangs after many rejects	High
D9	Renewal allowed while another member has reserved the book	Renewal logic	Reservation starvation; unfair service	Medium
D10	Date handling (time zone, leap year, month-end) wrong	Due-date calculation	Wrong due dates and fines	Medium
D11	Reference-only check placed after stock update or missing	N12 / N13 ordering	Reference books leave the library; stock mismatch	Medium

Combined impact: D1 and D2 together explain the observed spikes. D1 makes the CPU peak, and D2 means the memory never comes back down, so a long-running service eventually fails even though each single transaction is correct. D3 to D5 and D11 are business-rule defects that cause wrong data; D6 to D8 are robustness and security defects that can take the whole module down.

6. Business Rules and Decision Table

6.1 Reconstructed business rules

Rule	Statement
BR-1	Only an Active member (not suspended or expired) can borrow.
BR-2	A member can hold at most 5 books at one time.
BR-3	A member whose outstanding fine is above 100 cannot borrow.
BR-4	Only a book with an available copy can be issued; otherwise a reservation is offered.
BR-5	Reference-only books can never be issued.
BR-6	The loan period is 14 days; a renewal adds 14 days; at most 2 renewals.
BR-7	A book that is overdue or reserved by another member cannot be renewed.
BR-8	Fine is 2 per day after the due date, and cannot exceed the book price.

6.2 Decision table: Issue Book

Conditions / Actions	R1	R2	R3	R4	R5	R6	R7
C1: Member and book IDs found	Y	N	Y	Y	Y	Y	Y
C2: Member is Active	Y	-	N	Y	Y	Y	Y
C3: Books held < 5	Y	-	-	N	Y	Y	Y
C4: Fine <= 100	Y	-	-	-	N	Y	Y
C5: Copy available	Y	-	-	-	-	N	Y
C6: Not reference-only	Y	-	-	-	-	-	N
A1: Issue book, set due date	X						
A2: Reject with reason		X	X	X	X		X
A3: Offer reservation						X	
Message	Issued	Invalid ID	Not active	Limit reached	Clear fine	Reserve?	Reference

Y = true, N = false, "-" = not evaluated (earlier condition already decided the result). Each of the 7 rules becomes one test case (R1 = TC-P01, R2 = TC-N02/N06, R3 = TC-N03, R4 = TC-N04, R5 = TC-N05, R6 = TC-P07, R7 = TC-N07). These map to paths P10, P3/P7, P4, P5, P6, P8 and P9.

6.3 Decision table: Renew Book

Conditions / Actions	R1	R2	R3	R4
C1: Renewals < 2	Y	N	Y	Y
C2: Not overdue	Y	-	N	Y
C3: Not reserved by another member	Y	-	-	N
A1: Extend due date by 14 days	X			
A2: Reject with reason		X	X	X

6.4 Decision table: Fine calculation on return

Condition	Days late	Fine result
On time	0 or less	0 (negative values must not create a negative fine)
Late	1 to N	2 x days, capped at book price
Lost book	Not applicable	Book price plus processing charge; status = Lost
Corrupt date	Return date before issue date	Error; no fine, no stock change

7. System States, Events and Transitions

7.1 States of a book

Available, Reserved, Issued, Renewed, Overdue and Lost. Events are issue, reserve, return, renew, due date passed, reservation expiry, declared lost, and found/replaced.

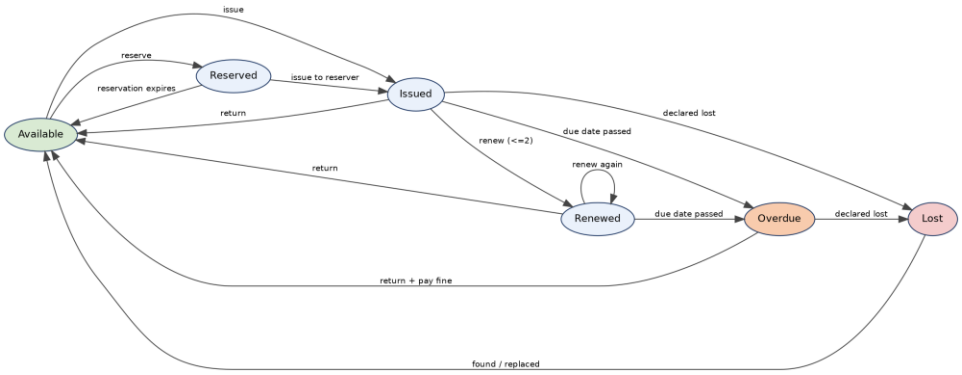


Figure 3: Book state transition diagram

7.2 Valid transitions

From state	Event	To state	Condition / action
Available	Issue	Issued	Member eligible (BR-1 to BR-5); due = +14 days
Available	Reserve	Reserved	Member eligible to reserve
Reserved	Issue to reserver	Issued	Only the member who reserved
Reserved	Reservation expires	Available	Hold time exceeded
Issued	Renew	Renewed	Renewals < 2, not reserved
Renewed	Renew again	Renewed	Only while count < 2
Issued / Renewed	Return	Available	Stock +1
Issued / Renewed	Due date passes	Overdue	Fine starts at 2 per day
Overdue	Return and pay fine	Available	Fine collected
Issued / Overdue	Declared lost	Lost	Charge price
Lost	Found / replaced	Available	Stock restored

7.3 Invalid transitions (negative state tests)

ID	From state	Event	Expected behaviour
ST-N1	Issued	Issue to another member	Rejected; offer reservation; state unchanged
ST-N2	Available	Return	Error "book not on loan"; stock unchanged
ST-N3	Available	Renew	Rejected; nothing to renew
ST-N4	Renewed (count = 2)	Renew	Rejected: renewal limit
ST-N5	Overdue	Renew	Rejected; ask to return and pay fine
ST-N6	Reserved	Issue to a different member	Rejected; only the reserver may borrow
ST-N7	Lost	Issue / Renew / Return	Rejected; only "found / replaced" allowed

State-transition coverage: every valid transition in 7.2 is tested once (0-switch coverage), then selected two-step sequences are tested (for example Available, Issued, Renewed, Overdue, Available) to catch defects that depend on history such as D9 and D10. Each invalid transition in 7.3 is attempted once and must leave the state and the stock count unchanged.

8. Summary

- **Reverse engineering:** CPU spikes S1 to S4 were mapped to Loops A, B and C and to the validation decision cluster N4 to N12 of the issue/return logic; the memory steps indicate retained data.
- **Path testing:** V(G) = 12, so 12 basis paths (P1 to P12) plus six resource-intensive scenarios (RIP-1 to RIP-6) target the heavy paths.

RUBRICS FOR CALCULATION:

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)	Marks
Understanding of System	20%	Demonstrates clear and in-depth understanding of the system/product and its functionality	Shows good understanding with minor gaps	Partial understanding; misses key aspects	Lacks understanding of the system/product	
Decomposition	25%	Effectively breaks down the system into components with detailed analysis	Decomposes system with minor gaps in analysis	Limited or incomplete decomposition	Unable to analyze or break down system	
Identification of Components	20%	Accurately identifies all components and explains their functions clearly	Identifies most components with minor errors	Identifies few components; explanations are weak	Unable to identify components or functions	
Reconstruction	20%	Successfully reconstructs or models the system with accuracy and logic	Reconstruction is mostly correct with minor issues	Partial reconstruction; lacks clarity	Unable to reconstruct or model system	
Documentation	15%	Well-organized, clear documentation with diagrams and structured explanation	Documentation is adequate with minor issues	Poorly organized or incomplete documentation	No proper documentation or presentation	